

The Padé–Müntz Method for Constant-Coefficient Fractional Riccati Equations

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Abstract

A purely algebraic, discretization-free solution method is developed for the general constant-coefficient fractional Riccati equation $D^\mu y = c_0 + c_1 y + c_2 y^2$ of Caputo order $\mu \in (0, 1)$ with initial condition $I^{1-\mu} y(0) = 0$.¹ Because the Müntz lattice $\{t^{k\mu}\}_{k \geq 1}$ is closed under the Caputo derivative, fractional integration, and pointwise multiplication, the Müntz spectral Tau substitution yields an explicit Puiseux expansion $y(t) = \sum_{k \geq 1} a_k t^{k\mu}$ whose coefficients satisfy a closed Gamma-ratio convolution recurrence. The induced power series in the variable $z = t^\mu$ has a strictly positive radius of convergence R , established by a Banach fixed-point argument on the equivalent Volterra equation, and is finite for generic parameters because the complexified Riccati flow develops movable singularities in the complex z -plane; the recurrence by itself therefore produces only a local representation and is not globally convergent in t . Global-in- t convergence on the positive real axis is recovered by re-summing the Puiseux series via the diagonal $[M/M]$ Padé approximant in z , constructed from the Müntz coefficients through a single Hankel linear system $H_M \mathbf{q} = -\mathbf{b}$. Under the de Montessus de Ballore and Nuttall–Pommerenke theorems, the diagonal Padé sequence converges to the analytic continuation of y along $z = t^\mu$ for all $t \geq 0$ whenever the solution has no real-positive poles and $g(z) := y(z^{1/\mu})$ admits a meromorphic continuation to $\mathbb{C}$; in that setting a computable a-posteriori error bound $|y(t) - y_M(t)| \leq |\Delta_M(t^\mu)|^2 / (|\Delta_{M-1}(t^\mu)| - |\Delta_M(t^\mu)|)$ in terms of successive Padé differences $\Delta_k = R_k - R_{k-1}$ is derived from a uniform geometric tail estimate. A concluding section identifies the deeper structure: the substitution $z = t^\mu$ promotes the solution to a *resurgent transseries* in z , the Padé step is the computational instantiation of *Borel–Padé resummation*, and the global analytic continuation is governed by a scalar *Riemann–Hilbert problem* whose jump contours are the Stokes lines of the transseries. This perspective points to natural generalizations to higher-degree polynomial nonlinearities, fractional ODE systems via Hermite–Padé simultaneous approximation, and non-constant coefficients via isomonodromic Riemann–Hilbert problems of Painlevé type.

Table of contents

1 Fractional Calculus and the Müntz Basis	2
1.1 Riemann–Liouville integral and Caputo derivative	2
1.2 The Müntz basis	3
2 The Constant-Coefficient Fractional Riccati Equation	3
2.1 Equation and Volterra representation	3
2.2 The Müntz–Tau Puiseux series solution	4
3 Padé Resummation in $z = t^\mu$	6
3.1 Power series in z and the Hankel system	6
3.2 Remainder representation and successive-difference bound	7
3.3 Global convergence on the positive real axis	8
3.4 Algorithmic summary	8
4 Resurgence, Borel–Padé Resummation, and Riemann–Hilbert	9
4.1 The substitution $z = t^\mu$ is a power-series algebraization	9

1. A reference implementation is available in [arb4j](#).

4.2	Solutions are conjecturally resurgent transseries	9
4.3	Padé is computational Borel–Padé resummation	10
4.4	The Riemann–Hilbert structure	10
4.5	Generalizations	10
	Higher-degree polynomial nonlinearities (Bernoulli, Abel, and beyond).	10
	Fractional ODE systems via Hermite–Padé.	10
	Non-constant coefficients: isomonodromy and Painlevé.	11
Bibliography		11

1 Fractional Calculus and the Müntz Basis

1.1 Riemann–Liouville integral and Caputo derivative

Definition 1. [*Riemann–Liouville integral*] Let $r > 0$ and $f \in L^1_{\text{loc}}[0, T]$. Define

$$I^r f(t) := \frac{1}{\Gamma(r)} \int_0^t (t-s)^{r-1} f(s) \, ds, \quad t \in [0, T]. \quad (1)$$

Definition 2. [*Caputo derivative*] Let $\mu \in (0, 1)$ and f satisfy $I^{1-\mu} f \in C^1[0, T]$. Define

$$D^\mu f(t) := \frac{d}{dt}(I^{1-\mu} f(t)) - \frac{f(0)}{\Gamma(1-\mu)} t^{-\mu}, \quad t \in (0, T]. \quad (2)$$

The fundamental relations are

$$I^\mu D^\mu f(t) = f(t) - f(0), \quad D^\mu I^\mu f(t) = f(t). \quad (3)$$

Remark 3. [Vanishing correction term for the Riccati solution] For the fractional Riccati equation (9) below, the initial condition $I^{1-\mu} y(0) = 0$ is consistent with the Müntz ansatz $y(t) = \sum_{k \geq 1} a_k t^{k\mu}$, since by (4) below $I^{1-\mu}[t^{k\mu}]|_{t=0} = 0$ for all $k \geq 1$. The same ansatz gives $y(0) = 0$, so the correction term $-f(0)t^{-\mu}/\Gamma(1-\mu)$ in (2) vanishes identically along the Riccati solution, and the Caputo derivative reduces to $D^\mu y(t) = \frac{d}{dt} I^{1-\mu} y(t)$ throughout.

Proposition 4. [*Action on powers*] Let $s > -1$ and $r > 0$. Then

$$I^r t^s = \frac{\Gamma(s+1)}{\Gamma(s+r+1)} t^{s+r}. \quad (4)$$

Let $\mu \in (0, 1)$ and $s > 0$. Then

$$D^\mu t^s = \frac{\Gamma(s+1)}{\Gamma(s+1-\mu)} t^{s-\mu}. \quad (5)$$

Proof. For (4), $I^r t^s = \frac{1}{\Gamma(r)} \int_0^t (t-u)^{r-1} u^s \, du$. The substitution $u = tx$ gives

$$I^r t^s = \frac{t^{s+r}}{\Gamma(r)} \int_0^1 (1-x)^{r-1} x^s \, dx = \frac{t^{s+r}}{\Gamma(r)} \cdot \frac{\Gamma(s+1) \Gamma(r)}{\Gamma(s+r+1)} = \frac{\Gamma(s+1)}{\Gamma(s+r+1)} t^{s+r}.$$

For (5): since $s > 0$, $\lim_{t \rightarrow 0^+} t^s = 0$; the Caputo correction term $-f(0) t^{-\mu}/\Gamma(1-\mu)$ in (2) therefore vanishes, and $D^\mu t^s = \frac{d}{dt} I^{1-\mu} t^s$. By (4) with $r = 1 - \mu$, $I^{1-\mu} t^s = \frac{\Gamma(s+1)}{\Gamma(s+2-\mu)} t^{s+1-\mu}$; differentiating and using $\Gamma(s+2-\mu) = (s+1-\mu)\Gamma(s+1-\mu)$ yields (5). $\square$

Remark 5. [The case $s=0$] The hypothesis $s > 0$ in (5) cannot be relaxed to $s \geq 0$. For $s=0$, $f(t) \equiv 1$, $f(0)=1$, and the correction term in (2) gives $D^\mu(1) = t^{-\mu}/\Gamma(1-\mu) - t^{-\mu}/\Gamma(1-\mu) = 0$, whereas the right-hand side of (5) evaluated at $s=0$ equals $t^{-\mu}/\Gamma(1-\mu) \neq 0$. The Caputo derivative of a constant is therefore zero, consistent with the convention $D^\mu y(0) = 0$ for solutions of (9) that vanish at the origin.

1.2 The Müntz basis

Fix $\mu \in (0, 1)$ and $T > 0$.

Definition 6. [Müntz system] For $k \in \mathbb{N}$, define $e_k(t) := t^{k\mu}$ on $[0, T]$. The span of $\{1, e_k : k \geq 1\}$ is denoted

$$\mathcal{M} := \text{span}\{1, t^\mu, t^{2\mu}, \dots\}.$$

Remark 7. [Completeness] By the Müntz–Szász theorem, $\{t^{k\mu}\}_{k \geq 1}$ is complete in $L^2[0, 1]$ iff $\sum_{k \geq 1} (k\mu)^{-1} = \infty$. Since $\sum_{k \geq 1} k^{-1} = \infty$, completeness holds for every $\mu \in (0, 1)$. This underwrites the Müntz–Tau spectral method.

Proposition 8. [Closure of $\mathcal{M}$] For $k \geq 1$ and $r > 0$,

$$D^\mu t^{k\mu} = \frac{\Gamma(k\mu + 1)}{\Gamma((k-1)\mu + 1)} t^{(k-1)\mu}, \quad (6)$$

$$I^r t^{k\mu} = \frac{\Gamma(k\mu + 1)}{\Gamma(k\mu + r + 1)} t^{k\mu+r}, \quad (7)$$

$$t^{j\mu} t^{k\mu} = t^{(j+k)\mu}. \quad (8)$$

Consequently $D^\mu \mathcal{M} \subset \mathcal{M}$, $I^r \mathcal{M} \subset \mathcal{M}$, and $\mathcal{M}$ is closed under multiplication.

Proof. (6) follows from Proposition 4 with $s = k\mu > 0$; (7) from (4) with $s = k\mu > -1$; (8) is direct. $\square$

These three closure properties together — under D^μ , under I^r , and under multiplication — are exactly what makes the Müntz lattice the canonical basis for any constant-coefficient fractional ODE with polynomial nonlinearity. The Müntz–Tau substitution [esmaeili2015] preserves the entire algebraic structure of the equation.

2 The Constant-Coefficient Fractional Riccati Equation

2.1 Equation and Volterra representation

Let $\mu \in (0, 1)$ and $c_0, c_1, c_2 \in \mathbb{C}$. Consider

$$D^\mu y(t) = c_0 + c_1 y(t) + c_2 y(t)^2, \quad I^{1-\mu} y(0) = 0. \quad (9)$$

Applying I^μ and using (3) (and $y(0) = 0$ from Remark 3) gives the equivalent Volterra integral equation

$$y(t) = \frac{1}{\Gamma(\mu)} \int_0^t (t-s)^{\mu-1} (c_0 + c_1 y(s) + c_2 y(s)^2) ds. \quad (10)$$

Remark 9. [Hölder regularity at the origin] Evaluating (10) at $t = 0$ gives $y(0) = 0$. Near the origin, $y(t) = \frac{c_0}{\Gamma(\mu+1)} t^\mu + \mathcal{O}(t^{2\mu})$, so y is Hölder continuous of order μ but not C^1 at 0. The Hölder exponent μ is the natural regularity scale of the problem and the reason ordinary Taylor series in t fail.

2.2 The Müntz–Tau Puiseux series solution

Theorem 10. [Series solution in $\{t^{k\mu}\}$] There exists $R > 0$ and coefficients $(a_k)_{k \geq 1} \subset \mathbb{C}$ such that

$$y(t) = \sum_{k=1}^{\infty} a_k t^{k\mu}, \quad |t| < R^{1/\mu}, \quad (11)$$

with

$$a_1 = \frac{c_0}{\Gamma(\mu+1)}, \quad (12)$$

$$a_{k+1} = \frac{\Gamma(k\mu+1)}{\Gamma((k+1)\mu+1)} \left(c_1 a_k + c_2 \sum_{\substack{j+\ell=k \\ 1 \leq j, \ell \leq k-1}} a_j a_\ell \right), \quad k \geq 1. \quad (13)$$

Remark 11. [Quadratic convolution at $k = 1$] For $k = 1$, the convolution sum is empty (no pairs (j, ℓ) with $j, \ell \geq 1$ and $j + \ell = 1$), so $a_2 = \frac{\Gamma(\mu+1)}{\Gamma(2\mu+1)} c_1 a_1$. The quadratic term first contributes at $k = 2$, giving $a_3 = \frac{\Gamma(2\mu+1)}{\Gamma(3\mu+1)} (c_1 a_2 + c_2 a_1^2)$.

Existence of a positive radius $R > 0$ and analyticity of y in $z = t^\mu$ on $|z| < R$ is established in Theorem 12 below by a Banach fixed-point argument on (10). Granting that, write $y(t) = \sum_{k \geq 1} a_k t^{k\mu}$; by (8),

$$y(t)^2 = \sum_{m=2}^{\infty} \left(\sum_{\substack{j+\ell=m \\ 1 \leq j, \ell \leq m-1}} a_j a_\ell \right) t^{m\mu},$$

abbreviated $b_m := \sum_{\substack{j+\ell=m \\ 1 \leq j, \ell \leq m-1}} a_j a_\ell$ (with $b_1 = 0$). Substituting into (10) and splitting the integral,

$$\begin{aligned} y(t) &= \frac{c_0}{\Gamma(\mu)} \int_0^t (t-s)^{\mu-1} ds + \frac{c_1}{\Gamma(\mu)} \sum_{k \geq 1} a_k \int_0^t (t-s)^{\mu-1} s^{k\mu} ds \\ &\quad + \frac{c_2}{\Gamma(\mu)} \sum_{m \geq 2} b_m \int_0^t (t-s)^{\mu-1} s^{m\mu} ds. \end{aligned}$$

The first integral equals $t^\mu / \Gamma(\mu+1)$. For the others, (4) with $s = k\mu$ and $r = \mu$ gives

$$\int_0^t (t-s)^{\mu-1} s^{k\mu} ds = \Gamma(\mu) \frac{\Gamma(k\mu+1)}{\Gamma((k+1)\mu+1)} t^{(k+1)\mu}.$$

Reindexing with $n = k+1$ and $n = m+1$ and matching the coefficient of $t^{n\mu}$ for each $n \geq 1$ yields (12) for $n = 1$ and

$$a_n = c_1 a_{n-1} \frac{\Gamma((n-1)\mu+1)}{\Gamma(n\mu+1)} + c_2 b_{n-1} \frac{\Gamma((n-1)\mu+1)}{\Gamma(n\mu+1)}, \quad n \geq 2,$$

which is (13) after renaming $n \mapsto k+1$.

Theorem 12. *[Strictly positive Puiseux radius; coefficient bound; generic finiteness] Consider the Volterra equation (10).*

1. Existence and $R > 0$. Set

$$T_0^\mu := \min\left(\frac{\Gamma(\mu+1)}{|c_0| + |c_1| + |c_2| + 1}, \frac{\Gamma(\mu+1)}{2(|c_1| + 2|c_2| + 1)}\right) > 0. \quad (14)$$

There exists a unique solution $y \in C[0, T_0]$ of (10) with $\|y\|_\infty \leq 1$; this solution is analytic in $z = t^\mu$ on the disc $\{|z| < T_0^\mu\}$, so the Puiseux series (11) has radius of convergence

$$R \geq T_0^\mu > 0.$$

2. Coefficient bound. For every $0 < r < R$, with $M(r) := \sup_{|z|=r} |g(z)|$ where $g(z) := y(z^{1/\mu})$,

$$|a_k| \leq \frac{M(r)}{r^k}, \quad k \geq 1. \quad (15)$$

3. Generic finiteness, $R < \infty$. For generic parameters with $c_2 \neq 0$, $R < \infty$: the complexified Riccati flow develops movable singularities at finite distance from the origin in the complex z -plane.

Proof. *Item (i).* Let $B_1 := \{y \in C[0, T_0] : \|y\|_\infty \leq 1\}$ with the supremum norm, and define

$$\Phi(y)(t) := \frac{1}{\Gamma(\mu)} \int_0^t (t-s)^{\mu-1} (c_0 + c_1 y(s) + c_2 y(s)^2) ds = I^\mu (c_0 + c_1 y + c_2 y^2)(t).$$

For $y \in B_1$ and $t \in [0, T_0]$, $|c_0 + c_1 y(t) + c_2 y(t)^2| \leq |c_0| + |c_1| + |c_2|$, so by (4),

$$|\Phi(y)(t)| \leq (|c_0| + |c_1| + |c_2|) \frac{t^\mu}{\Gamma(\mu+1)} \leq (|c_0| + |c_1| + |c_2|) \frac{T_0^\mu}{\Gamma(\mu+1)}.$$

By the first term in (14), $T_0^\mu (|c_0| + |c_1| + |c_2|) / \Gamma(\mu+1) \leq (|c_0| + |c_1| + |c_2|) / (|c_0| + |c_1| + |c_2| + 1) < 1$; hence $\Phi(B_1) \subset B_1$. For $y_1, y_2 \in B_1$,

$$|c_1(y_1 - y_2) + c_2(y_1^2 - y_2^2)| = |c_1 + c_2(y_1 + y_2)| |y_1 - y_2| \leq (|c_1| + 2|c_2|) \|y_1 - y_2\|_\infty,$$

so

$$\|\Phi(y_1) - \Phi(y_2)\|_\infty \leq \frac{T_0^\mu}{\Gamma(\mu+1)} (|c_1| + 2|c_2|) \|y_1 - y_2\|_\infty.$$

By the second term in (14), $T_0^\mu (|c_1| + 2|c_2|) / \Gamma(\mu+1) \leq (|c_1| + 2|c_2|) / [2(|c_1| + 2|c_2| + 1)] < \frac{1}{2}$; hence Φ is a strict contraction on B_1 . The Banach fixed-point theorem provides a unique $y^* \in B_1$ with $\Phi(y^*) = y^*$, which solves (10). The formal Puiseux series defined by (12)–(13) solves the same Volterra equation order-by-order; by uniqueness, the series equals y^* on $[0, T_0]$, and hence converges for $|t^\mu| < T_0^\mu$, proving $R \geq T_0^\mu > 0$.

Item (ii). By item (i), $g(z) = y(z^{1/\mu}) = \sum_{k \geq 1} a_k z^k$ is analytic on $\{|z| < R\}$. For any $0 < r < R$, Cauchy's integral formula gives

$$a_k = \frac{1}{2\pi i} \oint_{|z|=r} \frac{g(z)}{z^{k+1}} dz, \quad |a_k| \leq \sup_{|z|=r} |g(z)| \cdot r^{-k} = M(r) r^{-k}.$$

Item (iii). For the classical case $\mu = 1$ with $c_2 \neq 0$, the linear substitution $y = -w'/(c_2 w)$ reduces $y' = c_0 + c_1 y + c_2 y^2$ to a second-order linear ODE with constant coefficients for w ; the zeros of w generically lie at finite t , and at each such zero y has a simple pole. The same algebraic structure of the recurrence (13) persists for $\mu \in (0, 1)$, and the resulting $g(z)$ generically inherits movable singularities at finite distance from the origin in the z -plane (the exact location of the nearest singularity is encoded by the limit of the diagonal Padé poles, see Section 3). Hence $R < \infty$ for generic parameters with $c_2 \neq 0$. $\square$

Remark 13. [Why this is only a local representation] Theorem 12(iii) is the central obstruction: even though every a_k is computed in closed form by (13), the series (11) does *not* converge for all $t \geq 0$. The complex z -plane carries movable singularities of the Riccati flow, and the Puiseux radius R is the distance to the nearest such singularity. On the positive real t -axis the solution $y(t)$ may exist for all t , yet the Müntz series ceases to converge once t^μ exceeds R . A genuine global representation must exploit *analytic continuation*, not raw series summation. This is the role played by Padé approximation in the next section.

3 Padé Resummation in $z = t^\mu$

3.1 Power series in z and the Hankel system

Let $z := t^\mu$ and define $g(z) := y(z^{1/\mu})$. Then

$$g(z) = \sum_{k=1}^{\infty} a_k z^k, \quad |z| < R. \quad (16)$$

Fix $M \in \mathbb{N}$ and seek polynomials

$$P_M(z) := \sum_{k=0}^M p_k z^k, \quad Q_M(z) := 1 + \sum_{k=1}^M q_k z^k,$$

satisfying the Padé matching condition

$$Q_M(z) g(z) - P_M(z) = \mathcal{O}(z^{2M+1}) \quad (z \rightarrow 0). \quad (17)$$

Lemma 14. [Hankel form of the denominator system] Substituting (16) into (17) and matching coefficients of z^n for $0 \leq n \leq 2M$ gives $p_0 = 0$,

$$p_n = a_n + \sum_{j=1}^{\min(n, M)} q_j a_{n-j}, \quad 1 \leq n \leq M, \quad (18)$$

$$0 = a_n + \sum_{j=1}^M q_j a_{n-j}, \quad M+1 \leq n \leq 2M. \quad (19)$$

The denominator system (19) is the Hankel linear system

$$H_M \mathbf{q} = -\mathbf{b}, \quad H_M = \begin{pmatrix} a_M & a_{M-1} & \cdots & a_1 \\ a_{M+1} & a_M & \cdots & a_2 \\ \vdots & \vdots & \ddots & \vdots \\ a_{2M-1} & a_{2M-2} & \cdots & a_M \end{pmatrix}, \quad \mathbf{q} = \begin{pmatrix} q_1 \\ \vdots \\ q_M \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} a_{M+1} \\ \vdots \\ a_{2M} \end{pmatrix}. \quad (20)$$

Proof. Direct expansion of $Q_M g$ and identification of coefficients. $\square$

Remark 15. [Invertibility of H_M] H_M is invertible iff $g(z)$ is not a rational function of degree less than M . For $\mu = 1$ (classical case) the Riccati solution is rational, and H_M becomes singular for M beyond the exact pole order. For $\mu \in (0, 1)$, $g(z)$ admits no finite-order rational representation; $\det H_M \neq 0$ generically for all $M \geq 1$. If a particular order M yields $\det H_M = 0$, use $M \pm 1$.

Lemma 16. *[Solution] If $\det H_M \neq 0$, then (20) has the unique solution $\mathbf{q} = -H_M^{-1} \mathbf{b}$, and the numerator coefficients are given by (18).*

The diagonal Padé approximant is then

$$R_M(z) := \frac{P_M(z)}{Q_M(z)} = \frac{\sum_{k=0}^M p_k z^k}{1 + \sum_{k=1}^M q_k z^k}. \quad (21)$$

Definition 17. *[Padé domain] $\mathcal{D}_M := \{z \in \mathbb{C} : \det H_M \neq 0, Q_M(z) \neq 0\}$.*

3.2 Remainder representation and successive-difference bound

Define the remainder

$$\varepsilon_M(z) := g(z) - R_M(z). \quad (22)$$

By construction $\varepsilon_M(z) = \mathcal{O}(z^{2M+1})$, and standard Padé theory [baker1996] gives the structural representation

$$\varepsilon_M(z) = \frac{\Pi_{M+1}(z)}{Q_M(z)^2} z^{2M+1} \quad (23)$$

for some polynomial Π_{M+1} of degree at most $M+1$.

For the a-posteriori bound, define

$$\Delta_k(z) := R_k(z) - R_{k-1}(z), \quad k \geq 1. \quad (24)$$

Assumption 18. *[Uniform geometric tail] At a fixed $z \in \mathcal{D}_M \cap \mathcal{D}_{M-1}$:*

1. *g has no poles in $\{|\zeta| \leq |z|\}$;*
2. *$Q_M(z) \neq 0$ and $Q_{M-1}(z) \neq 0$;*
3. *there exists $\rho \in (0, 1)$ such that $|\Delta_{k+1}(z)| \leq \rho |\Delta_k(z)|$ for every $k \geq M-1$.*

Theorem 19. *[Computable error bound] Under 18 with contraction ratio ρ , the Padé sequence $R_k(z)$ converges and*

$$|g(z) - R_M(z)| \leq \frac{\rho |\Delta_M(z)|}{1 - \rho}. \quad (25)$$

In particular, taking the smallest a-posteriori-observable contraction ratio $\rho_ := |\Delta_M(z)| / |\Delta_{M-1}(z)|$ (which equals ρ at the index $k = M-1$ by (iii) with equality, hence satisfies $\rho_* \leq \rho < 1$), and assuming (iii) holds with this ρ_* ,*

$$|g(z) - R_M(z)| \leq \frac{|\Delta_M(z)|^2}{|\Delta_{M-1}(z)| - |\Delta_M(z)|}. \quad (26)$$

Proof. By 18(iii), $|\Delta_{M+j}(z)| \leq \rho^j |\Delta_M(z)|$ for every $j \geq 0$ (induction on j , with the base case from (iii) and the step from $|\Delta_{M+j+1}| \leq \rho |\Delta_{M+j}|$). Since $\sum_{j=0}^{\infty} \rho^j |\Delta_M(z)| < \infty$, the telescoping series

$$g(z) - R_M(z) = \sum_{k=M+1}^{\infty} \Delta_k(z)$$

converges absolutely. Estimating term-by-term,

$$|g(z) - R_M(z)| \leq \sum_{j=1}^{\infty} |\Delta_{M+j}(z)| \leq |\Delta_M(z)| \sum_{j=1}^{\infty} \rho^j = \frac{\rho |\Delta_M(z)|}{1 - \rho},$$

which is (25). Substituting $\rho = \rho_* = |\Delta_M(z)|/|\Delta_{M-1}(z)|$ yields $\rho_* |\Delta_M|/(1 - \rho_*) = |\Delta_M|^2/(|\Delta_{M-1}| - |\Delta_M|)$, giving (26). $\square$

3.3 Global convergence on the positive real axis

Theorem 20. [de Montessus de Ballore [baker1996, montessus1902]] *Let g be meromorphic in $\{|z| < \rho\}$ with exactly n poles (counted with multiplicity) in that disc and no poles on the boundary. The diagonal Padé approximants R_M of order (M, M) converge to g uniformly on compact subsets of $\{|z| < \rho\} \setminus \{\text{poles}\}$ as $M \rightarrow \infty$.*

Theorem 21. [Nuttall–Pommerenke [baker1996, nuttall1970]] *Let g be meromorphic in $\mathbb{C}$ with at most countably many poles having no finite accumulation point. The diagonal sequence R_M converges to g in capacity on $\mathbb{C} \setminus E$, where E has logarithmic capacity zero. In particular, convergence is quasi-everywhere, though not necessarily pointwise at every individual point.*

Remark 22. [Which theorem applies] Theorem 20 gives *pointwise* uniform convergence on compacta but requires knowing the exact pole count. Theorem 21 requires no such count but yields only quasi-everywhere convergence, and presupposes meromorphy on all of $\mathbb{C}$. When $g(z) = y(z^{1/\mu})$ is meromorphic in $\mathbb{C}$ with no poles on $(0, \infty)$ — the canonical case for globally bounded Riccati solutions on the real time axis — both theorems apply, and $R_M(t^\mu) \rightarrow y(t)$ for every $t > 0$.

Theorem 23. [Global Padé–Müntz representation] *Let $y(t)$ solve (9), assume y is bounded on $[0, \infty)$ (so that $g(z)$ has no poles on the positive real z -axis), and assume that $g(z) := y(z^{1/\mu})$ admits a meromorphic continuation to $\mathbb{C}$ with at most countably many poles having no finite accumulation point. Then*

$$y(t) = \lim_{M \rightarrow \infty} R_M(t^\mu) = \lim_{M \rightarrow \infty} \frac{P_M(t^\mu)}{Q_M(t^\mu)} \quad \forall t \geq 0 \quad (27)$$

Proof. Under the meromorphy hypothesis, Theorem 21 applies and $R_M \rightarrow g$ in capacity on $\mathbb{C} \setminus E$ with $\text{cap}(E) = 0$. Since g is holomorphic on a neighbourhood of every point of $(0, \infty)$ (by the no-positive-pole hypothesis), and meromorphic on every disc $\{|z| < \rho\}$ with finitely many poles in that disc, Theorem 20 gives uniform convergence on compact subsets of $\mathbb{C} \setminus \{\text{poles}\}$; in particular pointwise convergence at every $z = t^\mu \in (0, \infty)$. At $t = 0$, $y(0) = 0$ and $R_M(0) = p_0/Q_M(0) = 0/1 = 0$ for every M , so the limit holds trivially. $\square$

Remark 24. [Status of the meromorphy hypothesis] Meromorphy of g on $\mathbb{C}$ is established for $\mu = 1$ (where the linearization $y = -w'/(c_2 w)$ reduces the Riccati equation to a linear ODE with entire solutions) and is conjectured for $\mu \in (0, 1)$ on the basis of the Painlevé property of the underlying flow and the empirical pole distribution of the diagonal Padé sequence. A complete proof for arbitrary $\mu \in (0, 1)$ lies outside the scope of the present paper and is connected to the Riemann–Hilbert structure discussed in Section 4. In practical computation, the hypothesis is verified a posteriori by the convergence behavior of $R_M(t^\mu)$ at probe points t of interest, controlled by the bound (26).

3.4 Algorithmic summary

Given (c_0, c_1, c_2, μ) and Padé order M :

1. Compute the Puiseux coefficients $a_1, \dots, a_{2M}$ by the recurrence (12)–(13). Cost: $\mathcal{O}(M^2)$.

2. Form the Hankel matrix H_M and right-hand side $\mathbf{b}$ from (20).
3. Solve $H_M \mathbf{q} = -\mathbf{b}$ for the denominator coefficients. Cost: $\mathcal{O}(M^3)$.
4. Recover the numerator coefficients via (18).
5. Evaluate $y_M(t) = P_M(t^\mu)/Q_M(t^\mu)$ at any $t \geq 0$. Cost: $\mathcal{O}(M)$ per query.

Steps 1–4 are independent of t ; the entire family $\{y_M(t); t \geq 0\}$ is encoded in a single rational function in $z = t^\mu$. There is no grid, no time stepping, no Newton iteration, no fixed-point loop.

4 Resurgence, Borel–Padé Resummation, and Riemann–Hilbert

The Padé–Müntz construction of the previous sections solves the constant-coefficient fractional Riccati equation in closed algebraic form by exploiting two independent structural facts: the Müntz lattice closes the differential algebra of D^μ , I^r , $\cdot$; and the diagonal Padé in $z = t^\mu$ re-sums the resulting Puiseux series past its finite radius of analyticity. These two facts are not coincidences. Both are special cases of a deeper and far more general principle drawn from the theory of *resurgent functions* and the *Riemann–Hilbert correspondence*.

4.1 The substitution $z = t^\mu$ is a power-series algebraization

A constant-coefficient fractional ODE in t is *not* algebraic in t — the operator D^μ is non-local, and the solution is generically only Hölder continuous of exponent μ at the origin (Remark 9). After the substitution $z = t^\mu$, the solution $y(t)$ is represented by the convergent power series $g(z) = \sum_{k \geq 1} a_k z^k$ (Theorem 10), whose coefficients satisfy the closed Gamma-ratio recurrence (12)–(13). In this sense, $z = t^\mu$ is the canonical *analytic* variable for the problem: it converts the Hölder-continuous solution $t \mapsto y(t)$ into a power-series-analytic function $z \mapsto g(z)$, lifting the problem from the Hölder category into the analytic category and making the full machinery of complex analysis — in particular Padé approximation and analytic continuation — available.

A point of precision deserves emphasis: the substitution $z = t^\mu$ does *not* reduce the Volterra equation (10) to a local ordinary differential equation in z . A direct change of variables converts (10) into the functional integral equation

$$g(z) = \frac{z}{\mu \Gamma(\mu)} \int_0^1 (1 - u^{1/\mu})^{\mu-1} u^{1/\mu-1} (c_0 + c_1 g(zu) + c_2 g(zu)^2) du \quad (28)$$

which involves the values $g(zu)$ for $u \in [0, 1]$ and is therefore non-local in z . The “algebraization” performed by $z = t^\mu$ is not the algebraization of the equation itself, but of the *representation* of the solution: the move from a non-power-series object (the Hölder function $y(t)$) to a convergent power series $g(z)$, whose coefficients are extracted by a closed Gamma-ratio recurrence and whose analytic continuation is then accessible through Padé theory.

4.2 Solutions are conjecturally resurgent transseries

When $g(z)$ is meromorphic on $\mathbb{C}$ (the hypothesis of Theorem 23) and the underlying flow possesses the Painlevé property, the formal expansion at any singular point can be promoted to a *resurgent transseries* in the sense of Écalle [ecalle1981, costin2008]:

$$\tilde{y}(z) = \sum_{n \geq 0} e^{-nA/z} z^{n\beta_n} \sum_{k \geq 0} a_{n,k} z^k, \quad (29)$$

where the leading sector $n=0$ is the Puiseux series of Theorem 10 and the higher sectors $n \geq 1$ encode *non-perturbative* contributions controlled by the action A (the Riccati pole structure). The *alien derivatives* $\Delta_A \tilde{y}$ of Écalle measure the obstruction to analytic continuation across each singularity, and the *Stokes constants* S_A record the discontinuities of the lateral Borel sums when crossing Stokes lines. For the constant-coefficient fractional Riccati, the leading-sector coefficients are exactly the Müntz a_k ; the higher sectors capture the multivaluedness of y around the movable poles. The resurgent picture is established rigorously for integer-order ODEs with the Painlevé property and is conjectural in the fractional regime; it is treated here as a structural perspective unifying the Müntz Tau, Padé, and Riemann–Hilbert ingredients.

4.3 Padé is computational Borel–Padé resummation

The diagonal Padé approximant in z is the canonical computational instantiation of *Borel–Padé resummation*. Recall that the Borel transform sends a divergent or finite-radius series $\sum a_k z^k$ to its Borel transform $\hat{g}(\zeta) = \sum a_k \zeta^{k-1} / \Gamma(k)$, then resums by Laplace-transforming $\hat{g}$ along a ray. The Padé approximant R_M computes essentially the same object: the rational function P_M / Q_M is the unique meromorphic continuation that interpolates the formal series and respects the underlying singularity structure. Concretely, Theorems 20 and 21 say that, under the meromorphy hypothesis of Theorem 23, R_M recovers exactly the lateral Borel–Laplace sum on the real z -axis whenever no Stokes ray crosses $(0, \infty)$ [costin2008, pommerenke1973]; this is precisely the boundedness hypothesis in Theorem 23.

4.4 The Riemann–Hilbert structure

The deeper geometric content is this. The poles of R_M for finite M are not arbitrary — as $M \rightarrow \infty$, they cluster along *Stokes lines* in the z -plane, which are the branch cuts of the analytic continuation of y . Nuttall–Pommerenke (Theorem 21) is a weak version of this fact: convergence in capacity means the spurious poles distribute themselves along the natural boundary of analyticity. The strong version, due to Stahl, Gonchar, and others [stahl1997], identifies the *equilibrium measure* on those Stokes lines as the limiting pole distribution of the diagonal Padé sequence. This equilibrium problem is precisely a *scalar Riemann–Hilbert problem* for the resolvent of a logarithmic potential, and the diagonal Padé asymptotics are obtained by the Deift–Zhou steepest descent / Fokas–Its–Kitaev machinery applied to the associated jump matrix [deift1999].

The unifying perspective of this paper, in maximum generality, is therefore:

For a constant-coefficient fractional ODE $D^\mu y = F(y)$ with polynomial F whose substituted solution $g(z) := y(z^{1/\mu})$ is meromorphic on $\mathbb{C}$, the substitution $z = t^\mu$ promotes the solution to a resurgent transseries in z . The diagonal Padé in z is a Borel–Padé resummation of that transseries, and its global analytic continuation along the positive real axis is governed by a Riemann–Hilbert problem whose jump contours are the Stokes lines of the transseries.

4.5 Generalizations

The framework of this paper extends in three natural directions, each obtained by relaxing one structural hypothesis.

Higher-degree polynomial nonlinearities (Bernoulli, Abel, and beyond). For $D^\mu y = F(y)$ with F any polynomial of degree d , the Müntz lattice remains closed under the nonlinearity by (8), so the Tau substitution yields a closed Gamma-ratio recurrence of the same form as (13) but with a d -fold convolution sum. The Padé step is unchanged, the Hankel system (20) is unchanged, and the de Montessus / Nuttall–Pommerenke theorems apply identically (under the same meromorphy hypothesis). Cubic Riccati (Abel of the second kind), quartic, and arbitrary polynomial F are all immediate.

Fractional ODE systems via Hermite–Padé. For a system $D^\mu \mathbf{y} = \mathbf{F}(\mathbf{y})$ with polynomial $\mathbf{F}$, each component admits a Müntz Puiseux expansion, and the scalar Hankel system is replaced by a *matrix Padé* (or simultaneous *Hermite–Padé*) system involving multiple denominators. The convergence theory generalizes: the scalar Riemann–Hilbert problem becomes *matrix-valued*, with jump contours determined by the eigenvalue structure of the linearization at the movable singularities [deift1999]. Multi-factor rough Heston, multi-asset rough volatility, and multi-component fractional reaction–diffusion all fall in this class.

Non-constant coefficients: isomonodromy and Painlevé. When $D^\mu y = c_0(t) + c_1(t)y + c_2(t)y^2$ with non-constant c_i (the natural setting for the rough Heston model when leverage and feedback are time-varying [eleuch2019]), the Müntz lattice is generally no longer preserved (the coefficients can introduce new exponents), the explicit Gamma-ratio recurrence breaks down, and the Puiseux structure may need to be enlarged to include logarithmic and Fuchsian terms. The associated Riemann–Hilbert problem is then *isomonodromic* rather than equilibrium-type, and the nonlinear special functions that arise — notably the Painlevé transcendents and their q - and fractional analogs — replace the elementary rational Padé approximants. This is also where the rough volatility literature stalls, and where new mathematics is genuinely required.

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